
Nise 2.13

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Use MATLAB to generate the following transfer function two different ways

Ratio of Factors

```
% Array of zeros
z = [-15 -26 -72];
% Array of poles
p = [0 -55 roots([1 5 30])' -56 roots([1 27 52])'];
% Gain
k = 5;
Gzpk = zpk(z,p,k)
```

Gzpk =

$$\frac{5 (s+15) (s+26) (s+72)}{s (s+55) (s+56) (s+24.91) (s+2.087) (s^2 + 5s + 30)}$$

Continuous-time zero/pole/gain model.

Ratio of Polynomials

Numerator

```
num = 5*conv([1 15],conv([1 26],[1 72]));
% Denominator
a = [1 0];
b = [1 55];
c = [1 5 30];
d = [1 56];
e = [1 27 52];
den = conv(a,conv(b,conv(c,conv(d,e))));
Gtf = tf(num,den)
```

Gtf =

$$5 s^3 + 565 s^2 + 16710 s + 140400$$

$$s^7 + 143 s^6 + 6849 s^5 + 123717 s^4 + 788690 s^3 + 3.469e06 s^2$$

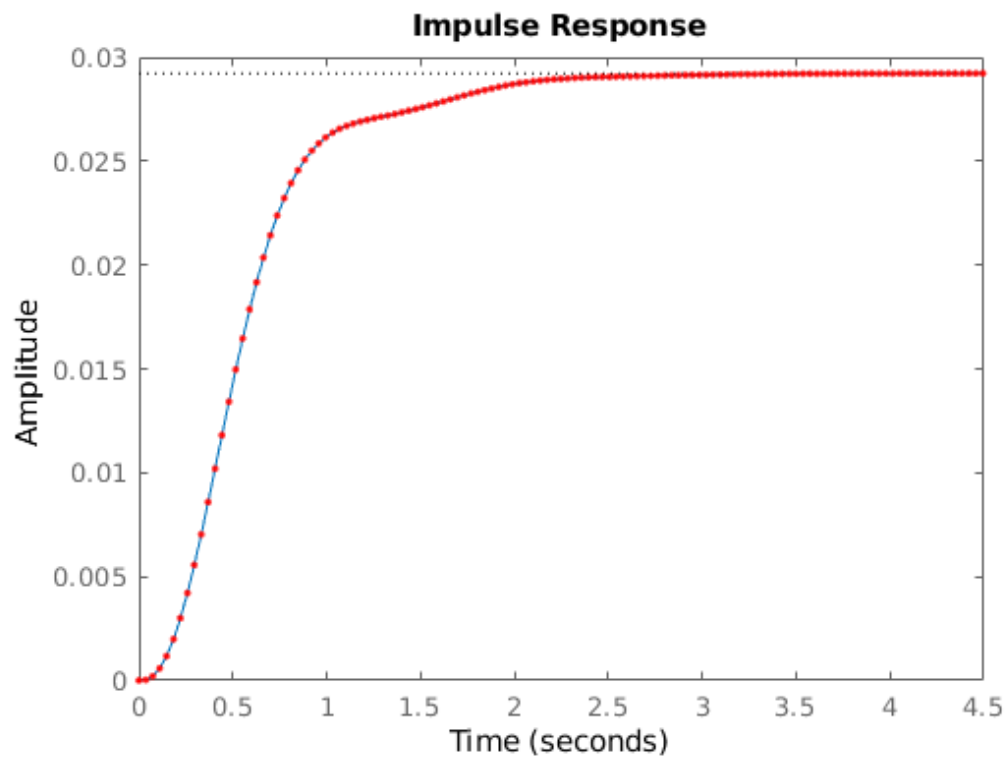
$$4.805e06 s \quad +$$

Continuous-time transfer function.

Optional

Test by asking MATLAB to plot the impulse response and making sure they yield the same dynamics.

```
figure(1);
clf();
impz(Gzpk)
hold on
impz(Gtft, 'r.')
```



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